

Appendix: scoped morphological view

Dataset `loss_mitigation_forms` **Component** `risk-pooling` **Mechanism** `MC1 = pooling`
24 forms × 9 characteristics. This view is restricted to one declared component set. Comparative claims never run on the full characteristic set: with enough characteristics any two forms separate, so a matrix printed over all of them reports how many characteristics were used rather than anything about the forms. At this n the matrix below is a coverage map and not evidence; the comparative reading is confined to Table 2, whose two characteristics were fixed before the coding.

Table 1: Scoped matrix. States are coded; see the character-state list. Missingness is never collapsed: - no row entered, .NR not recorded in the source, .IL illegible, .NA inapplicable, 0 an observed absence.

form	CN1	HZ1	HZ2	MB3	MC1	PR1	PY0	PY1	PY2
<code>averia_pool</code>	0	1	1	1	2	0	0	.NA	.NA
<code>confraternity_fund_it</code>	0	2	1	0	2	0	1	0	.NR
<code>craft_pension_edinburgh</code>	0	.NR	-	1	2	0	1	3	4
<code>craft_widows_fund_edinburgh</code>	0	.NR	-	0	2	P	1	1	.NR
<code>friendly_society_england</code>	0	.NR	P	0	2	0	1	0	3
<code>friendly_society_female_england</code>	0	.NR	P	0	2	P	1	0	3
<code>general_average</code>	2	2	P	1	2	0	1	0	3
<code>guild_box_brabant</code>	0	.NR	-	1	2	0	1	0	3
<code>guild_box_dutch</code>	0	2	P	1	2	0	1	0	3
<code>guild_relief_england</code>	.NR	.NR	-	0	2	0	1	3	4
<code>ko_daimanin_1828</code>	-	-	-	-	2	-	-	-	.NR
<code>ko_hokinosawa_1820</code>	0	.NR	-	0	2	0	2	2	0
<code>ko_hokoin_1832</code>	0	.NR	-	0	2	0	2	2	.NR
<code>ko_mitarai_gin_1805</code>	0	.NR	-	0	2	0	2	2	.NR
<code>ko_mitarai_mujin_1773</code>	0	.NR	-	0	2	0	2	2	0
<code>ko_mochiyori_yutsu_1876</code>	0	.NR	-	0	2	0	2	2	2
<code>ko_shijunin_1829</code>	0	-	-	-	2	-	-	-	0
<code>ko_usui_bango_1822</code>	0	.NR	-	0	2	0	2	2	.NR
<code>ko_yamakuni_1555</code>	0	.NR	-	.NR	2	0	2	2	.NR
<code>tontine</code>	-	-	-	-	2	.NR	-	-	-
<code>tontine_en_1693</code>	1	0	-	0	2	0	2	1	.NR
<code>tontine_fr_royal</code>	-	0	-	0	2	P	2	-	-
<code>torinoke_mujin</code>	0	.NR	-	0	2	0	2	2	0
<code>widows_fund_scotland</code>	-	0	-	1	2	1	1	1	-

Character states

CN1 *contribution timing* — 0 = ex-ante-periodic; 1 = ex-ante-lump; 2 = ex-post-assessment

HZ1 *hazard correlation* — 0 = idiosyncratic; 1 = covariate; 2 = mixed

HZ2 *hazard responsiveness* — ternary: 1 present, P partial, 0 absent

MB3 *participation basis* — 0 = voluntary; 1 = compulsory

MC1 *mitigation mechanism* — 0 = allocation; 1 = spreading; 2 = pooling

PR1 *peril priced ex ante* — ternary: 1 present, P partial, 0 absent

PY0 *pool output* — 0 = collective-good; 1 = individual-indemnity; 2 = individual-draw

Table 2: The comparative reading, printed with full values and no state codes. PR1 takes both values *inside* MC1=pooling at a fixed date, so ex-ante pricing is a separate axis rather than a stage of mechanism; PY0 separates the forms on whether a draw corresponds to the recipient's own realised loss.

form	PR1 peril priced ex ante	PY0 pool output
averia_pool	0	collective-good
confraternity_fund_it	0	individual-indemnity
craft_pension_edinburgh	0	individual-indemnity
craft_widows_fund_edinburgh	P	individual-indemnity
friendly_society_england	0	individual-indemnity
friendly_society_female_england	P	individual-indemnity
general_average	0	individual-indemnity
guild_box_brabant	0	individual-indemnity
guild_box_dutch	0	individual-indemnity
guild_relief_england	0	individual-indemnity
ko_daimanin_1828	-	-
ko_hokinosawa_1820	0	individual-draw
ko_hokoin_1832	0	individual-draw
ko_mitarai_gin_1805	0	individual-draw
ko_mitarai_mujin_1773	0	individual-draw
ko_mochiyori_yutsu_1876	0	individual-draw
ko_shijunin_1829	-	-
ko_usui_bango_1822	0	individual-draw
ko_yamakuni_1555	0	individual-draw
tontine	.NR	-
tontine_en_1693	0	individual-draw
tontine_fr_royal	P	individual-draw
torinoke_mujin	0	individual-draw
widows_fund_scotland	1	individual-indemnity

PY1 *payout trigger* — 0 = realised-loss; 1 = life-event; 2 = rotation; 3 = need-assessed
PY2 *allocation rule* — 0 = rotation-lot; 1 = rotation-fixed; 2 = auction; 3 = indemnity; 4 = need-assessed

Forms

averia_pool Avería (compulsory convoy levy) *Spanish / Carrera de Indias, 1521-1681*

confraternity_fund_it Confraternity and guild mutual support - Italian *Latin Christendom / Italian, 11c-15c*

craft_pension_edinburgh Craft pensions and supply, incorporation of Mary's Chapel *Scottish / Edinburgh, 1670-1768*

craft_widows_fund_edinburgh Widows' fund annuity scheme, incorporation of Mary's Chapel *Scottish / Edinburgh, 1768-*

friendly_society_england Friendly society / box club *English, 1870-1914 (this coding)*

friendly_society_female_england Female friendly society *English, 1780-1830 (this coding)*

general_average General average (jettison contribution) *Mediterranean / lex Rhodia, antiquity onward*

guild_box_brabant Ambachtsbus / armenbus — craft guild sickness box *Low Countries / Brabant, 1250-1600*

guild_box_dutch Gildebus — guild sickness, burial, old-age and widows fund *Dutch Republic, 1550-1800*

guild_relief_england English religious gild relief *English, c.1350-1400*

ko_daimanin_1828 ??? (???) *Japanese / Kawachi, 1828-1833+*

ko_hokinosawa_1820 ????? (????, ???) *Japanese / Echigo, 1820-1838*

ko_hokoin_1832 ????? (the kō joined by Hōkō-in) *Japanese / Dewa-Yamagata, 1832-*

ko_mitarai_gin_1805 ?????(???) *Japanese / Aki, 1805-*

ko_mitarai_mujin_1773 ???(???, ????? ????) *Japanese / Aki, 1773-1785*

ko_mochiyori_yutsu_1876 ????? (Mochiyori Yūtsū-sha) *Japanese / Kawachi-Settsu, 1876-1884*

ko_shijunin_1829 ??? (???) *Japanese / Kawachi, 1829-1843*

ko_usui_bango_1822 ????? (???) *Japanese / Kawachi, 1822-*

ko_yamakuni_1555 ????? (?????, ? ?????) *Japanese / Tanba, 1555-1563*

tontine Tontine *European, 17c onward*

tontine_en_1693 Tontine — English (Million Act 5 & 6 Will. & Mar. c. 5) *English, 1693-1789*

tontine_fr_royal Tontine — French royal series (fourteen age classes) *French, 1689-1759*

torinoke_mujin ????? (winner released from further contribution) *Japanese, 17c-1930s*

widows_fund_scotland Scottish Ministers' Widows' Fund *Scottish, 1744 onward*

All codings marked **coder=ai** are provisional and pending human verification. Cells carrying **[verify]** in **source_ref** are asserted against an unpinned citation.